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CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY CHANGA

B.Tech. (CE/EE/IT) Sem - I Examination, 2009-10

ME-102 : Fundamentals of Mechanical Engineering

Date: 15.12.09, Tuesday,

Time: 2:00 pm to 5:00 pm

Maximum Marks:70

Instructions:

1. Attempt all the questions.
2. Figures to the right indicate full marks.
3. Make suitable assumptions and draw neat figures wherever if required.

SECTION I

- Q:1 (a). Explain Boyle's law [02]
(b). Differentiate: gas and vapour. [03]
(c). Define work and power. Also, mention similarities between heat and work. [05]

- Q:2 (a). For adiabatic process, prove that $PV^\gamma = C$ [05]
(b). Establish equations for polytropic process: (i). Work done and (ii). Heat transfer. [05]

OR

- Q:2 (a). Derive the equation $PV = mRT$ for perfect gas. [05]
(b). Heat is supplied to 1 kg. of air at 40 bar and 900 K at constant pressure till the volume is doubled. The air is then expanded adiabatically till the Pressure drops to 1 bar. Calculate work done and heat transferred During (i). Constant pressure process and (ii). Adiabatic process. Take $R = 287 \text{ J/KgK}$, $C_v = 0.7 \text{ KJ/KgK}$, $\gamma = 1.4$ for air. [05]

- Q:3 (a). Explain the working of four Stroke diesel engine with neat diagram. [05]
(b). Derive the equation for air standard efficiency (η) of Otto cycle. [05]
(c). A 4-stroke cycle petrol engine has 120 mm bore and stroke is 1.25 times bore. It consumes 4.5 Kg of fuel per hour having calorific value of 41100 KJ/Kg. If the engine speed is 800 r.p.m. The mean effective pressure is $770 \times 10^3 \text{ N/m}^2$. Find indicated thermal efficiency. [05]

OR

- Q:3 (a). i. Draw skeleton of simple I. C. engine and label different components. [02]
ii. A car having a four stroke diesel engine with compression ratio 21:1 and expansion ratio 10.5:1. Find the cut-off ratio and air standard efficiency. [03]
(b). i. Differentiate brake and clutch. [02]
ii. Advantages of gear drive over belt drive. [03]
(c). Classify belt and pulley drive. [05]

SECTION II

- Q:4 (a). Draw enthalpy-temperature diagram for steam formation. [02]
 (b). Calculate the amount of heat required to convert 4 kg of water at 20 °C in to steam at 8 bar and 200 °C. Take $C_{PS}=2.1$ KJ/kg-K and $C_{P(water)}=4.187$ KJ/kg-k [03]
 (c). Explain separating & throttling calorimeter with neat sketch. [05]

- Q:5 (a). What is stoichiometric air requirement? [05]
 A coal sample gave the following analysis by weight:
 Carbon 86%, hydrogen 5%, sulphur 0.8%, oxygen 6% and remainder being incombustible. Determine the minimum air required per kg of coal for chemically correct composition.
 (b). Draw the neat sketch of Cochern boiler and explain its working. [04]
 (c). Define following terms: [06]
 (i). Refrigeration (ii). Air conditioning (iii). C.O.P (iv). Degree of superheat (v). Latent heat of evaporation (vi). Entropy

OR

- Q:5 (a). i. State advantages disadvantages of gaseous fuels. [03]
 ii. Give comparison of LPG & CNG. [02]
 (b). Enlist the different mountings of boiler. Draw neat sketch of water level indicator & state its function and location. [04]
 (c). i. Define the term 'equivalent evaporation' and explain its significance. [02]
 ii. In boiler trial, the following observations were made: [04]

Duration of trial=	24 hours
Steam pressure =	7 bar
Coal consumption=	1200 kg
Mass of water evaporated=	12000 kg
Feed water temperature=	50°C
Calorific value of fuel=	32000 KJ/Kg
Dryness fraction of steam=	1

Calculate: (i) Equivalent evaporation (ii) Boiler efficiency

- Q:6 (a). Define 'ton of refrigeration'. Explain Vapour compression Refrigeration system. [05]
 (b). Define volumetric efficiency with P-V diagram and prove that it is given as: [05]

$$\eta_v = 1 - C \left[\left(\frac{P_2}{P_1} \right)^{\frac{1}{n}} - 1 \right]$$

OR

- Q:6 (a). Classify the pumps. Explain the construction and working of centrifugal pump with neat sketch. [05]
 (b). A Single cylinder, single stage, single acting reciprocating compressor runs at 300 r.p.m. and takes 1 m³ of air per minute at 1.013 bar and 15 °C and delivers it at 7 bar. Assuming that the law of compression is $PV^{1.35} = C$ and clearance is negligible. Calculate (i) Indicated power (ii) cylinder bore required assuming stroke to bore ratio of 5:1. [05]

-Best of Luck-